Checkpoint 6 Maths — Practice Test 3 — Mark Scheme

Paper 1: 45 minutes · no calculator · Paper 2: 45 minutes · calculator allowed · Total 80 marks

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 45 minutes — 40 marks

1 At the school fun fair there is a "guess the mass" game. The cake weighs 3.586 kg. **[1]**

What is the value of the digit 8 in 3.586?

- ☐ A) 8 ones
- ☐ B) 8 tenths
- ☐ C) 8 hundredths
- ☐ D) 8 thousandths

Answer: C) 8 hundredths
Notes / working: 3.586 = 3 ones + 5 tenths + **8 hundredths** + 6 thousandths. The 8 is in the second place after the decimal point, so its value is 8 hundredths (0.08). Answer **C**.

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2 A farm keeps its milk and meat in cold rooms. **[1]**

(a) The meat room is at −7 °C. The temperature rises by 12 degrees. What is the new temperature?

Answer: 5 °C
Notes / working: Start at −7 and count on 12: 7 steps take you to 0, then 5 more steps take you to **5 °C**. (−7 + 12 = 5)

(b) The milk room is at 4 °C. The freezer is at −15 °C. What is the difference between the two temperatures? **[1]**

Answer: 19 degrees
Notes / working: From −15 up to 0 is 15 degrees, and from 0 up to 4 is 4 more degrees. 15 + 4 = **19 degrees**. Common mistake: 15 − 4 = 11 (this ignores that the temperatures are on opposite sides of zero).

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3 Karim counts back in equal steps. **[1]**
1.4, 0.9, 0.4, _____, −0.6

Write the missing number.

Answer: -0.1
Notes / working: Each step is 0.5 less: 1.4 − 0.5 = 0.9, 0.9 − 0.5 = 0.4, 0.4 − 0.5 = **−0.1**, and −0.1 − 0.5 = −0.6 ✓. Common mistake: writing 0.1 (forgetting that the count has gone past zero).

4 A bookshop orders 24 boxes of bookmarks. Each box has 1 367 bookmarks. **[2]**

How many bookmarks does the shop order altogether? Show your working.

Answer: 32808
Notes / working: Estimate: 1 400 × 24 ≈ 33 600, so the answer is a little less than this.
1 367 × 20 = 27 340
1 367 × 4 = 5 468
27 340 + 5 468 = **32 808**
Mark scheme: M1 for a complete correct method (e.g. 1 367 × 20 and 1 367 × 4 seen, or a correct grid/column layout) with no more than one arithmetic slip; A1 for 32 808.

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5 A farmer collects 756 eggs. She packs them in trays. Each tray holds 18 eggs. **[2]**

How many trays does she fill? Show your working.

Answer: 42 trays
Notes / working: Estimate: 760 ÷ 20 ≈ 38, so the answer is about 40.
Chunking: 40 × 18 = 720. 756 − 720 = 36 = 2 × 18. 40 + 2 = **42 trays**.
Check: 42 × 18 = 420 + 336 = 756 ✓
Mark scheme: M1 for a complete correct method (e.g. 40 × 18 = 720 and 36 ÷ 18 = 2, or a correct long-division layout); A1 for 42.

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6 **[1]**

(a) Which calculation has the answer 29?

- ☐ A) 4 + 6 × 3 − 1
- ☐ B) (4 + 6) × 3 − 1
- ☐ C) 4 + 6 × (3 − 1)
- ☐ D) (4 + 6) × (3 − 1)

Answer: B) (4 + 6) × 3 − 1
Notes / working: A: 4 + 18 − 1 = 21. **B: 10 × 3 − 1 = 30 − 1 = 29** ✓. C: 4 + 6 × 2 = 4 + 12 = 16. D: 10 × 2 = 20. Brackets are worked out first, then multiplication, then addition and subtraction. Answer **B**.

(b) Work out 50 − 8 × 5 **[1]**

Answer: 10
Notes / working: Multiply first: 8 × 5 = 40. Then 50 − 40 = **10**. Common mistake: working left to right, 50 − 8 = 42, 42 × 5 = 210.

7 Rana makes prize bags for the fun fair. She has 24 red balloons and 36 blue balloons. **[1]**
Every bag must have the same number of red balloons and the same number of blue balloons. No balloons are left over.

What is the greatest number of bags she can make?

Answer: 12 bags
Notes / working: The number of bags must be a **common factor** of 24 and 36.
Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24
Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36
The greatest common factor is **12**, so 12 bags (each with 2 red and 3 blue balloons).

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8 **[1]**

(a) Which number is divisible by 9?

- ☐ A) 2 571
- ☐ B) 5 162
- ☐ C) 7 235
- ☐ D) 3 474

Answer: D) 3 474
Notes / working: A number is divisible by 9 when its digit sum is divisible by 9.
2 571 → 15 ×, 5 162 → 14 ×, 7 235 → 17 ×, **3 474** → **3 + 4 + 7 + 4 = 18** ✓ (18 is a multiple of 9).
Answer **D**.

(b) Tarek says, "Every number that is divisible by 3 is also divisible by 6." **[1]**
Give an example to show that Tarek is wrong.

Answer: 1 mark: any odd multiple of 3, e.g. 3, 9, 15, 21, 27 (it may be with or without a reason such as "15 ÷ 3 = 5 but 15 is odd so it is not divisible by 6"). Do not accept an even multiple of 3 (e.g. 12, 18) or a number that is not a multiple of 3.
Notes / working: A number is divisible by 6 only when it is divisible by **both 2 and 3**. So any **odd** multiple of 3 breaks Tarek's rule. Example: **15** — digit sum 6, so it is divisible by 3, but it is odd, so it is not divisible by 6.

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9 **[1]**

Write the cube number that is between 100 and 130.

Answer: 125
Notes / working: Cube numbers: 1 × 1 × 1 = 1, 8, 27, 64, **5 × 5 × 5 = 125**, 216 … Only 125 is between 100 and 130.

10 Amir builds model cars in a workshop. Each car needs 4 wheels. He always orders 3 spare wheels as well. **[1]**

The rule for his order is:
number of wheels = 4 × number of cars + 3

(a) How many wheels does he order for 9 cars?

> **Answer:** 39 wheels
> **Notes / working:** 4 × 9 + 3 = 36 + 3 = **39 wheels**.

(b) Mona says, "Amir ordered exactly 50 wheels." **[1]**
Explain why Mona must be wrong.

> **Answer:** 1 mark: a correct reason, e.g. "50 − 3 = 47 and 47 is not a multiple of 4 / 47 ÷ 4 is not a whole number", or "the totals go 7, 11, 15, … 47, 51, so 50 is not in the list", or "4 × cars is even, add 3 gives an odd number, but 50 is even". Do not accept "50 is too big" or an answer with no reason.
> **Notes / working:** Work backwards: 50 − 3 = 47, and 47 ÷ 4 = 11 r 3, which is not a whole number of cars. (Also: 4 × a whole number is always even, and even + 3 is always **odd**, but 50 is even.) So 50 can never be an order total.

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11 A model car is 100 times shorter than the real car. The real car is 4.2 m long. **[1]**

How long is the model car, in metres?

> **Answer:** 0.042 m
> **Notes / working:** 4.2 ÷ 100: every digit moves two places to the right. 4.2 → 0.42 → **0.042 m** (that is 4.2 cm). Common mistake: 0.42 (only one place moved).

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12 An axle for a model car is 6.24 cm long. **[1]**

Complete: 6.24 = 6 + 0.1 + _____

> **Answer:** 0.14
> **Notes / working:** 6.24 − 6 = 0.24, and 0.24 − 0.1 = **0.14**. Check: 6 + 0.1 + 0.14 = 6.24 ✓. (0.24 is 2 tenths and 4 hundredths; regroup it as 1 tenth and 14 hundredths.)

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13 A farmer shares 7 bales of hay equally between 4 horses. **[1]**

How many bales does each horse get? Give your answer as a fraction or a mixed number.

> **Answer:** 1.75 or 7/4 or 1 3/4 bales
> **Notes / working:** 7 ÷ 4 = 7/4 = **1 3/4 bales** (each horse gets 1 whole bale and 3 quarters of another). A fraction is the numerator divided by the denominator.

14 On the farm, 18 of the 24 cows are black and white. **[1]**

What fraction of the cows are black and white? Give your answer in its simplest form.

Answer: 3/4
Notes / working: 18/24. The highest common factor of 18 and 24 is 6. 18 ÷ 6 = 3 and 24 ÷ 6 = 4, so **3/4**. (9/12 is equivalent but not in its simplest form.)

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15 A farmer plants 1/3 of a field with wheat and 1/4 of the field with maize. The rest of the field is grass. **[2]**

What fraction of the field is grass? Show your working.

Answer: 5/12
Notes / working: Use the common denominator 12: 1/3 = 4/12 and 1/4 = 3/12.
Wheat and maize: 4/12 + 3/12 = 7/12
Grass: 12/12 − 7/12 = **5/12**
Mark scheme: M1 for 7/12 seen (or 1/3 and 1/4 both changed correctly to twelfths); A1 for 5/12 oe.

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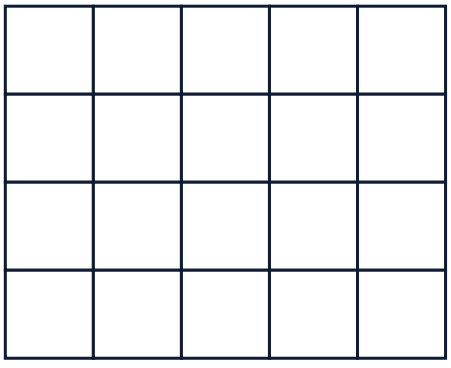
16 A bag of rabbit food holds 3/4 kg. It is shared equally between 3 rabbits. **[1]**

How much food does each rabbit get, in kilograms?

Answer: 1/4 or 0.25 kg
Notes / working: 3/4 ÷ 3: three quarters shared into 3 equal parts gives one quarter each. **1/4 kg** (= 0.25 kg, or 250 g).

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17 A bookshop is making a sale poster on this grid. **[1]**



Shade 35% of the grid. (Copy the grid on paper.)

Answer: 1 mark: exactly 7 squares shaded (in any position). Do not accept 35 squares or any other number of squares.
Notes / working: The grid has 4 × 5 = 20 squares. 100% = 20 squares, so 5% = 1 square and 35% = 7 × 5% = **7 squares**.

18 **[1]**

(a) Write <, > or = to make this correct:
3/4 ☐ 0.8

> **Answer:** <
> **Notes / working:** 3/4 = 0.75 and 0.75 is less than 0.80, so **3/4 < 0.8**.

(b) Sara says, "0.45 is less than 2/5." **[1]**
Is Sara correct? Explain how you know.

> **Answer:** 1 mark: "No" with a correct comparison, e.g. 2/5 = 0.4 (or 0.40) and 0.45 > 0.40; or 2/5 = 40% and 0.45 = 45%; or 2/5 = 8/20 and 0.45 = 9/20. Do not accept "No" with no reason or a wrong reason.
> **Notes / working: No.** 2/5 = 4/10 = 0.40, and 0.45 > 0.40. So 0.45 is **greater** than 2/5. (In percentages: 45% > 40%.)

19 At the swimming gala, Laila swims 50 m in 31.25 seconds. Nour swims 50 m in 29.8 seconds. **[1]**

How many seconds faster is Nour than Laila?

> **Answer:** 1.45 seconds
> **Notes / working:** Line up the decimal points: 31.25 − 29.80 = **1.45 seconds**. Common mistake: lining up the last digits (31.25 − 2.98).

20 A notebook in the bookshop costs $2.35. **[1]**

How much do 6 notebooks cost?

> **Answer:** 14.1 $
> **Notes / working:** 2.35 × 6: 2 × 6 = 12, 0.35 × 6 = 2.10. 12 + 2.10 = **$14.10**. Estimate check: about 2 × 6 = 12 to 3 × 6 = 18 ✓.

21 The swimming gala lasts 3.25 hours. **[1]**

Complete: 3.25 hours = 3 hours and _____ minutes

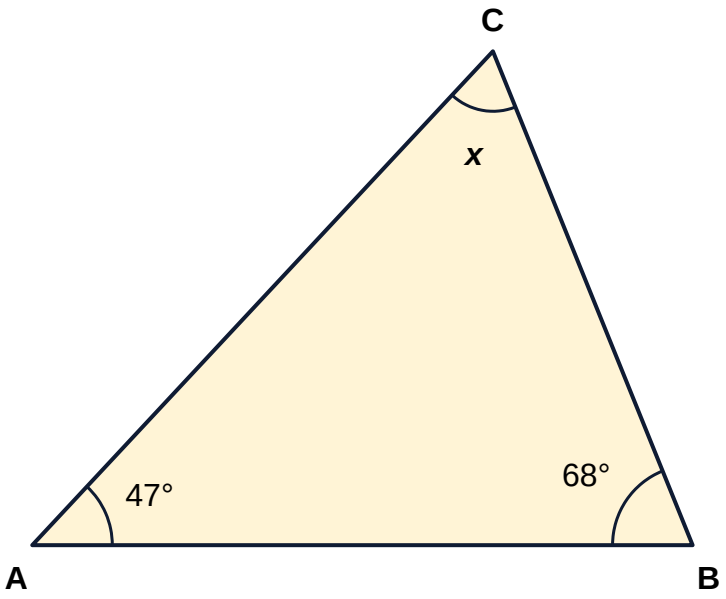
> **Answer:** 15 minutes
> **Notes / working:** 0.25 hours = 1/4 of an hour = 60 ÷ 4 = **15 minutes**. Common mistake: 25 minutes (time is not a decimal system — 1 hour is 60 minutes, not 100).

22 Amir wants a paper wheel for a model car. **[1]**

Use compasses to draw a circle with a **diameter** of 7 cm. Mark the centre and draw one radius. Label it "radius".

Answer: 1 mark: a circle with radius 3.5 cm (diameter between 6.8 cm and 7.2 cm) AND the centre marked AND a straight line from the centre to the circumference labelled radius. Do not accept a circle of radius 7 cm.
Notes / working: The radius is half the diameter: 7 ÷ 2 = **3.5 cm**. Open the compasses to 3.5 cm on a ruler, put the point on the centre and turn once. A radius goes from the centre to the circumference (the edge).

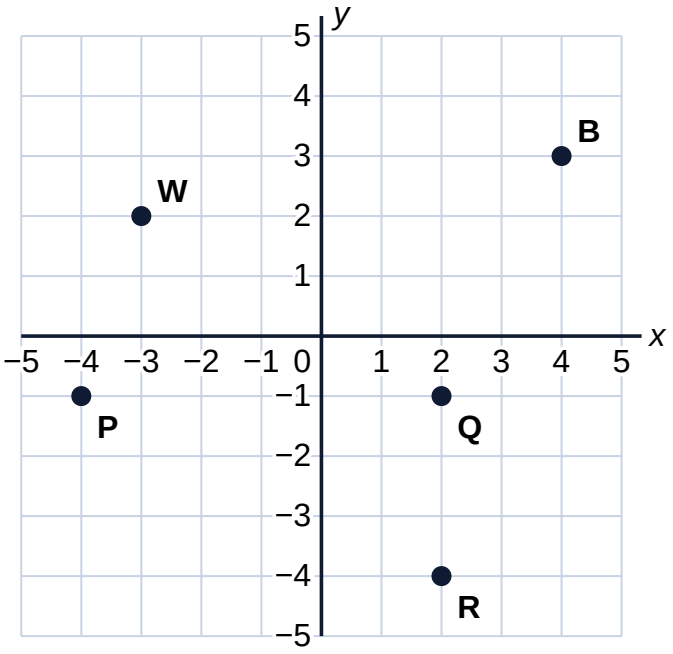
23 A triangular flag hangs at the fun fair. The diagram is not drawn to scale. **[1]**



Work out the size of angle x.

Answer: 65 °
Notes / working: The angles in a triangle add up to 180°. 47 + 68 = 115. 180 − 115 = **65°**.

24 This map of a farm is drawn on a coordinate grid. W is the well and B is the barn. **[1]**



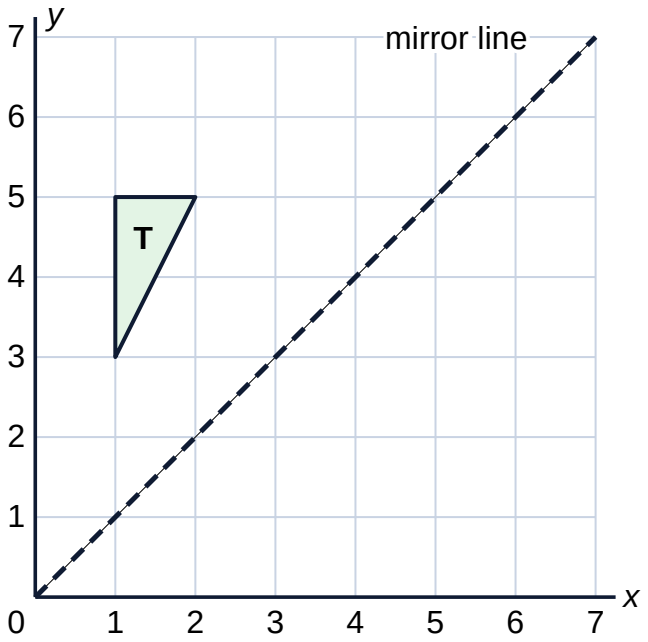
(a) Write the coordinates of the well, W.

Answer: (-3, 2) or (−3, 2) or (-3,2) or -3, 2 or −3, 2
Notes / working: From the origin go 3 left (x = −3) then 2 up (y = 2): **(−3, 2)**. The x-coordinate (across) always comes first.

(b) P, Q and R are three corners of a rectangular field. Write the coordinates of the fourth corner. **[1]**

Answer: (-4, -4) or (−4, −4) or (-4,-4) or -4, -4 or −4, −4
Notes / working: P(−4, −1) and Q(2, −1) form the top side; R(2, −4) is below Q. The fourth corner is below P, level with R: x from P and y from R → **(−4, −4)**.

25 The diagram shows a triangle T and a mirror line. **[1]**



Draw the reflection of triangle T in the mirror line. Write the coordinates of the corners of your reflection. (Copy the grid on paper.)

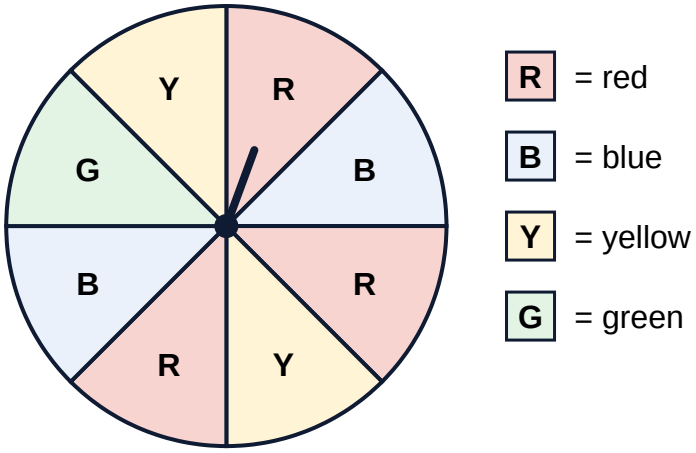
Answer: 1 mark: reflection drawn with corners at (3, 1), (5, 1) and (5, 2). Do not accept a translation or a reflection in a vertical/horizontal line.
Notes / working: Each point moves straight across the diagonal mirror line to the same distance on the other side. For this mirror line the x- and y-coordinates swap: (1, 3) → **(3, 1)**, (1, 5) → **(5, 1)**, (2, 5) → **(5, 2)**. Check: each corner and its image are the same number of diagonal steps from the line.

26 **[1]**

Huda says, "A square is a special kind of rhombus."
Is Huda correct? Explain your answer.

Answer: 1 mark: "Yes" with a reason, e.g. a rhombus has four equal sides (and opposite sides parallel) and a square also has four equal sides; or a square is a rhombus whose angles are all right angles. Do not accept "Yes" with no reason, or a reason about right angles only.
Notes / working: Yes. A rhombus is any quadrilateral with four equal sides (its opposite sides are parallel and its diagonals cross at right angles). A square has all of these properties — it is a rhombus that also has four right angles.

27 This spinner is used for a game of chance at the fun fair. All 8 parts are the same size. **[1]**



(a) Which word best describes the chance of the spinner landing on red?

- ☐ A) unlikely
- ☐ B) even chance
- ☐ C) likely
- ☐ D) certain

Answer: A) unlikely
Notes / working: 3 of the 8 equal parts are red. 3/8 is less than 1/2 (4/8), so landing on red is **unlikely**. Answer **A**.

(b) Can "landing on red" and "landing on blue" happen at the same time on one spin? Write the name for events like these. **[1]**

Answer: 1 mark: "No" (the pointer can only stop on one part) AND the word mutually exclusive. Accept recognisable misspellings. Do not accept "no" on its own.
Notes / working: No — the spinner stops on one part, and a part cannot be red and blue. Events that cannot happen at the same time are called **mutually exclusive**.

28 Five swimmers in Team A swim one length. Their times are: **[1]**

Swimmer	Ali	Omar	Hany	Ziad	Adam
Time (seconds)	41	38	45	39	43

(a) Find the median time.

Answer: 41 seconds
Notes / working: Put the times in order: 38, 39, **41**, 43, 45. The middle (3rd) value is **41 seconds**. Common mistake: taking the middle of the unsorted list (45).

(b) The range of Team A's times is 7 seconds. Team B's times have a range of 12 seconds. Which team's times are more consistent (closer together)? Give a reason. **[1]**

> **Answer:** 1 mark: Team A AND a reason based on the range, e.g. "Team A has the smaller range (7 < 12), so its times are closer together". Do not accept Team A with no reason.
> **Notes / working: Team A.** The range measures how spread out the data are. 45 − 38 = 7 s is smaller than 12 s, so Team A's times are closer together (more consistent).

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29 The farm records how many eggs are collected on six days: **[1]**
12, 15, 18, 27, 30, 36

Copy and complete the Carroll diagram by writing each number in the correct box.

	Multiple of 9	**Not a multiple of 9**
Even		
Not even		

> **Answer:** 1 mark for all six correct: Even & multiple of 9: 18, 36; Even & not a multiple of 9: 12, 30; Not even & multiple of 9: 27; Not even & not a multiple of 9: 15.
> **Notes / working:** Multiples of 9 (digit sum 9, 18 …): 18, 27, 36. Even numbers: 12, 18, 30, 36.

	Multiple of 9	**Not a multiple of 9**
Even	18, 36	12, 30
Not even	27	15

Paper 2 — 45 minutes — 40 marks

1 At the swimming gala, Yasmin's time is 34.76 seconds. **[1]**

Round 34.76 to the nearest tenth of a second.

Answer: 34.8 seconds
Notes / working: Look at the hundredths digit: 6 is 5 or more, so round the tenths up: 34.7 → **34.8 s**. Common mistake: 34.7 (cutting off instead of rounding) or 35 (nearest whole number).

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2 A sign in a bookshop says: "3/10 off all atlases". **[1]**

Which of these is the same as 3/10?

- ☐ A) 3%
- ☐ B) 0.03
- ☐ C) 30%
- ☐ D) 0.13

Answer: C) 30%
Notes / working: 3/10 = 0.3 = 30/100 = **30%**. Answer **C**. (3% = 0.03 = 3/100.)

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3 In the bookshop sale, a dictionary normally costs $24. The sale price is 30% less. **[2]**

Work out the sale price of the dictionary. Show your working.

Answer: 16.8 $
Notes / working: 10% of $24 = $2.40, so 30% = 3 × 2.40 = $7.20.
Sale price = 24 − 7.20 = **$16.80**
(Or: 70% of 24 = 0.7 × 24 = 16.80.)
Mark scheme: M1 for 7.2(0) seen or 0.7 × 24 (70% of 24); A1 for 16.8(0).

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4 Last month the bookshop sold 36 atlases. This month it sold 5/4 of that number. **[1]**

How many atlases did it sell this month?

Answer: 45 atlases
Notes / working: 5/4 of 36: 36 ÷ 4 = 9, and 9 × 5 = **45**. An improper fraction as an operator makes the amount bigger (5/4 = 1 1/4).

5 The bookshop's machine prints 4 896 bookmarks each day for 31 days. **[1]**

Which is the best estimate of the total number of bookmarks printed?

- ☐ A) 150 000
- ☐ B) 15 000
- ☐ C) 1 500 000
- ☐ D) 1 500

Answer: A) 150 000
Notes / working: Round: 4 896 ≈ 5 000 and 31 ≈ 30. 5 000 × 30 = **150 000**. Answer **A**. (Exact: 4 896 × 31 = 151 776, which is close ✓.)

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6 In the model-car workshop, 3 model cars need 45 ml of paint. **[1]**

(a) How much paint is needed for 7 model cars?

Answer: 105 ml
Notes / working: 1 car needs 45 ÷ 3 = 15 ml. 7 cars need 7 × 15 = **105 ml**.

(b) A tin holds 250 ml of paint. How many model cars can be painted completely with one tin? **[1]**

Answer: 16 cars
Notes / working: 250 ÷ 15 = 16.66… The 17th car cannot be finished, so **16 cars** (16 × 15 = 240 ml; 10 ml is left over). Common mistake: rounding up to 17.

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7 On a farm, for every 2 cows there are 7 chickens. **[1]**

(a) There are 63 chickens. How many cows are there?

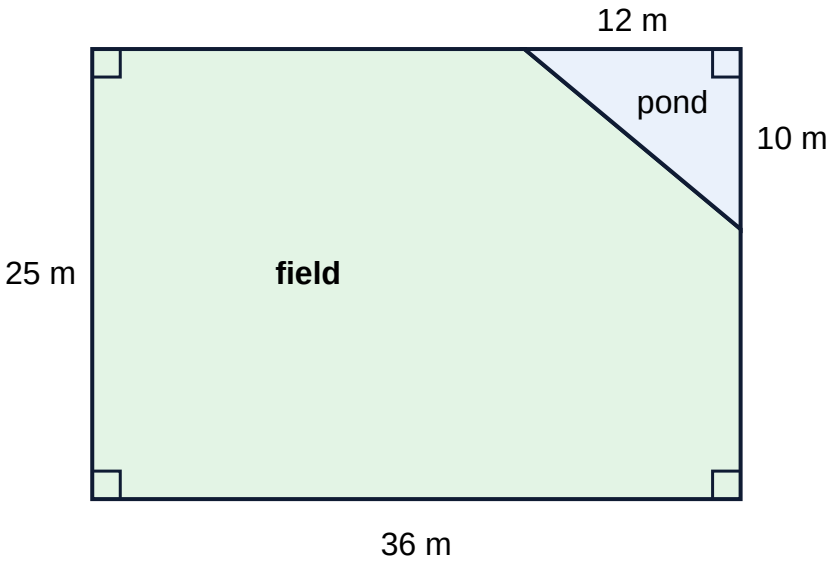
Answer: 18 cows
Notes / working: cows : chickens = 2 : 7. 63 ÷ 7 = 9, so multiply both parts by 9: 2 × 9 = **18 cows** (ratio 18 : 63).

(b) Which ratio is equivalent to 2 : 7? **[1]**

- ☐ A) 4 : 9
- ☐ B) 7 : 2
- ☐ C) 6 : 20
- ☐ D) 8 : 28

Answer: D) 8 : 28
Notes / working: Multiply both parts by the same number: 2 × 4 = 8 and 7 × 4 = 28, so **8 : 28** ✓. Answer **D**. (4 : 9 adds 2 to each part — not equivalent; 7 : 2 is the wrong order; 6 : 20 would need 6 : 21.)

8 A field is a rectangle 36 m long and 25 m wide. One corner of it is a pond in the shape of a right-angled triangle. The diagram is not drawn to scale. **[1]**



(a) Work out the area of the pond.

> **Answer:** 60 m^2
> **Notes / working:** A right-angled triangle is half of a rectangle: 12 × 10 = 120, 120 ÷ 2 = **60 m²**.

(b) Work out the area of the field that is **not** pond. Show your working. **[2]**

> **Answer:** 840 m^2
> **Notes / working:** Whole rectangle: 36 × 25 = 900 m². Take away the pond: 900 − 60 = **840 m²**. Mark scheme: M1 for 36 × 25 (= 900) seen, or 900 − their (a); A1 for 840 (FT their (a): 900 − their pond area).

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9 At the gala, Ali's time for one race is 1.6 minutes. **[1]**

Complete: 1.6 minutes = 1 minute and _____ seconds

> **Answer:** 36 seconds
> **Notes / working:** 0.6 of a minute = 0.6 × 60 = **36 seconds**. Common mistake: 1 minute 6 seconds or 1 minute 60 seconds.

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10 Eight swimmers swim 50 m backstroke. Their times, in seconds, are: **[1]**
32.4, 30.9, 33.5, 31.2, 30.9, 34.0, 31.8, 32.1

(a) Find the range of the times.

> **Answer:** 3.1 seconds
> **Notes / working:** Range = largest − smallest = 34.0 − 30.9 = **3.1 seconds**.

(b) Calculate the mean time. Show your working. **[2]**

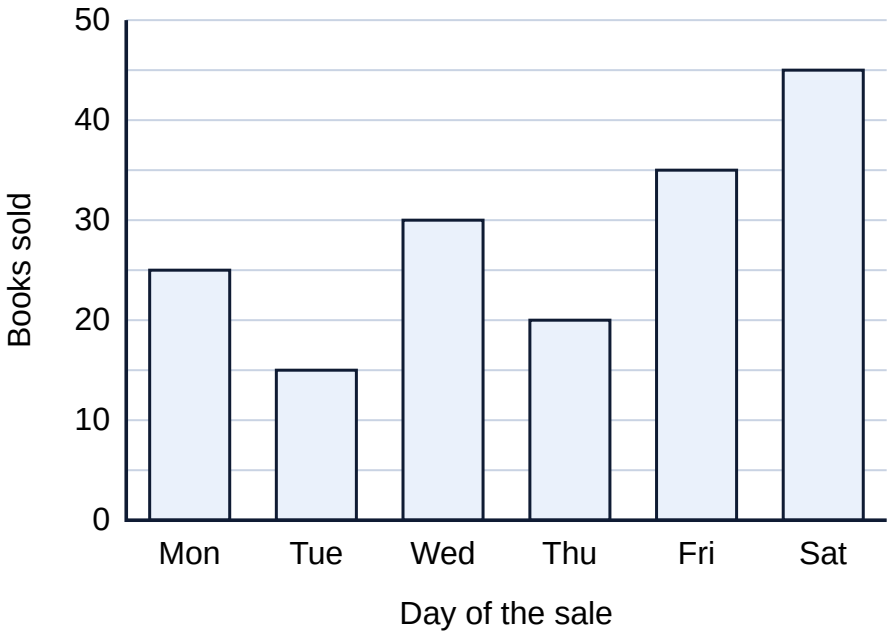
> **Answer:** 32.1 seconds
> **Notes / working:** Total: 32.4 + 30.9 + 33.5 + 31.2 + 30.9 + 34.0 + 31.8 + 32.1 = 256.8
> Mean = 256.8 ÷ 8 = **32.1 seconds**
> Mark scheme: M1 for 256.8 seen or (sum of the 8 times) ÷ 8; A1 for 32.1.

(c) The coach says, "The mode is the best average to describe these times." **[1]**
Do you agree? Give a reason.

> **Answer:** 1 mark: "No" with a valid reason, e.g. the mode (30.9 s) is the fastest/smallest time so it is not typical; only two swimmers have the same time; the mean (32.1) or median (31.95) is closer to the middle of the data. Do not accept "No" with no reason.
> **Notes / working: No.** The mode is 30.9 s, but that is the **fastest** time in the list — only two swimmers happen to share it. The mean (32.1 s) or the median ((31.8 + 32.1) ÷ 2 = 31.95 s) describes a typical swimmer better, because these times are all different decimals and a repeat is just chance.

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11 The bar chart shows the number of books a bookshop sold on each day of its sale. **[1]**



(a) How many more books were sold on Saturday than on Tuesday?

> **Answer:** 30 books
> **Notes / working:** Saturday = 45 (halfway between 40 and 50), Tuesday = 15 (halfway between 10 and 20). 45 − 15 = **30 books**.

(b) Before the sale, the manager predicted, "We will sell more books every day than the day before." **[1]**
Was the prediction correct? Use the bar chart to explain.

> **Answer:** 1 mark: "No" with evidence from the chart, e.g. Tuesday (15) was less than Monday (25), or Thursday (20) was less than Wednesday (30). Do not accept "No" without a reason, or "Yes".
> **Notes / working: No.** Sales went down on Tuesday (25 → 15) and again on Thursday (30 → 20). Overall the sales rose by Saturday, but not every day.

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12 Lina wants to find out if model cars with bigger wheels roll further after going down a ramp. **[1]**

(a) Which of these is continuous data?

- ☐ A) the colour of each car
- ☐ B) the distance each car rolls
- ☐ C) the number of wheels on each car
- ☐ D) the name of each car's maker

> **Answer:** B) the distance each car rolls
> **Notes / working:** Continuous data are **measured** and can take any value in a range (e.g. 2.35 m). Distance is measured, so it is continuous. Colour and maker's name are categorical; number of wheels is counted, so it is discrete. Answer **B**.

(b) Write the **two** sets of data Lina needs to collect for each car to answer her question. **[1]**

> **Answer:** 1 mark for both: the size of the wheels (e.g. wheel diameter/radius) AND the distance the car rolls. Do not accept the colour/mass of the car or "the time" in place of one of these.
> **Notes / working:** Her question links **wheel size** and **distance rolled**, so she must measure both for each car — for example the diameter of the wheels (cm) and the distance from the bottom of the ramp to where the car stops (cm or m).

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13 At the fun fair, there is a "Lucky Duck" game. You pick a duck; some ducks win a prize. **[1]**
On the first morning, 50 people played and 12 won.
By the end of the week, 500 people had played and 140 had won.

(a) Use the results for the whole week to estimate the probability of winning. Give your answer as a fraction or a decimal.

> **Answer:** 0.28 or 7/25 or 140/500 or 14/50
> **Notes / working:** Probability ≈ wins ÷ games = 140 ÷ 500 = **0.28** (= 28/100 = 7/25).

(b) Explain why the results for the whole week give a better estimate than the results for the first morning. **[1]**

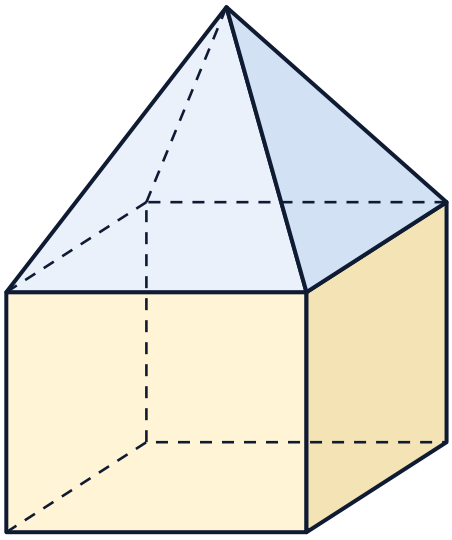
> **Answer:** 1 mark: idea that more trials (500 is more than 50) give a more reliable estimate / results settle down with more trials / a small number of trials can be affected by chance. Do not accept "because it is a week" with no idea of more trials.
> **Notes / working:** The week has **500 trials**, but the morning has only 50. With a small number of trials, chance can easily push the result up or down (12 ÷ 50 = 0.24). With many more trials the results settle closer to the true probability, so the estimate is more reliable.

(c) Next week, about 2 000 people will play. About how many people would you expect to win? **[1]**

> **Answer:** 560 people
> **Notes / working:** 0.28 × 2 000 = **560 people** (or 140 wins in 500 games → 4 × 140 = 560 in 2 000 games). FT 2 000 × their (a).

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14 Amir makes a model garage. The diagram shows its shape. **[1]**



Which two 3D shapes make the model garage?

- ☐ A) a cube and a triangular prism
- ☐ B) a cylinder and a cone
- ☐ C) a cuboid and a square-based pyramid
- ☐ D) a cuboid and a triangular prism

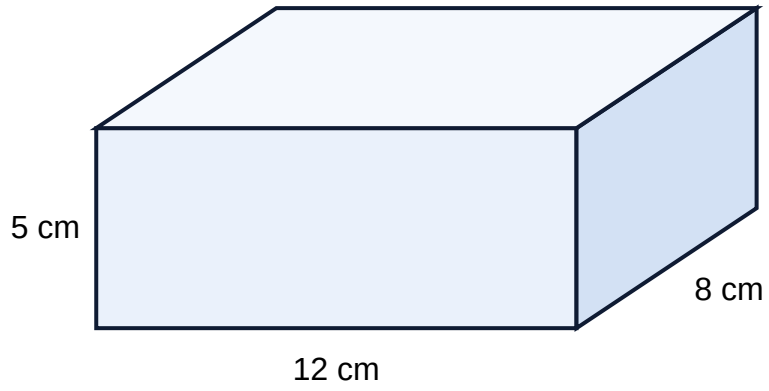
> **Answer:** C) a cuboid and a square-based pyramid
> **Notes / working:** The bottom part has six rectangular faces — a **cuboid**. The top part has a square base and four triangular faces meeting at one point (the apex) — a **square-based pyramid**. Answer **C**. (A triangular prism would have a ridge along the top, not a point.)

15 **[1]**

Sketch a net of a square-based pyramid. (Draw it on paper.)

Answer: 1 mark: a net with one square and four triangles, each triangle joined to a different side of the square (the cross/star shape), or any other arrangement that folds into a square-based pyramid. Do not accept a net with 5 triangles, with two squares, or with two triangles on the same side of the square.
Notes / working: A square-based pyramid has **1 square face and 4 triangular faces**. The easiest net: draw the square, then draw one triangle on each of its four sides, all pointing outwards. When the triangles fold up, their tips meet at the apex.

16 A model-car kit comes in a closed cardboard box. The box is a cuboid 12 cm long, 8 cm wide and 5 cm high. **[2]**



Work out the total surface area of the box. Show your working.

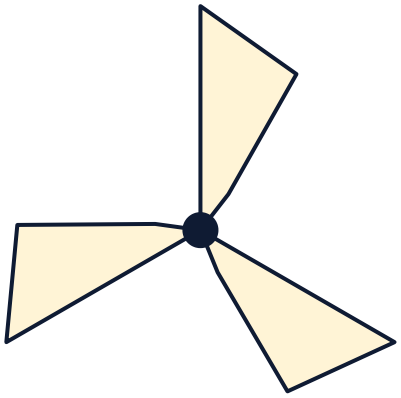
Answer: 392 cm^2
Notes / working: A cuboid has three pairs of equal rectangular faces:
top and bottom: 2 × (12 × 8) = 2 × 96 = 192
front and back: 2 × (12 × 5) = 2 × 60 = 120
the two ends: 2 × (8 × 5) = 2 × 40 = 80
Total = 192 + 120 + 80 = **392 cm^2**
Mark scheme: M1 for at least two of 96, 60, 40 (or 192, 120, 80) seen; A1 for 392. Common mistake: 12 × 8 × 5 = 480 — that is the volume (cm^3), not the surface area.

17 A bottle of model paint says "capacity 250 ml". **[1]**
Ravi says, "The capacity of the bottle is the same as the volume of the bottle."

Explain the difference between the capacity and the volume of the bottle.

Answer: 1 mark: capacity is the amount (of liquid) the bottle can hold (the space inside); volume is the amount of space the bottle itself takes up (including its walls/material). Accept equivalent wording. Do not accept an answer that only describes one of them.
Notes / working: Capacity = how much the bottle can hold (the space inside it, here 250 ml).
Volume = how much space the whole bottle takes up, including the plastic or glass walls — so its volume is a bit more than its capacity.

18 This is the logo of the fun-fair windmill ride. **[1]**



What is the order of rotational symmetry of the logo?

Answer: 3
Notes / working: Turn the logo about its centre. It looks the same after 1/3 of a turn (120°), 2/3 of a turn and a full turn — **3 times** in one full turn, so it has rotational symmetry of **order 3**.

19 **[1]**

Use a protractor to draw an angle of 125°. Write what type of angle it is.

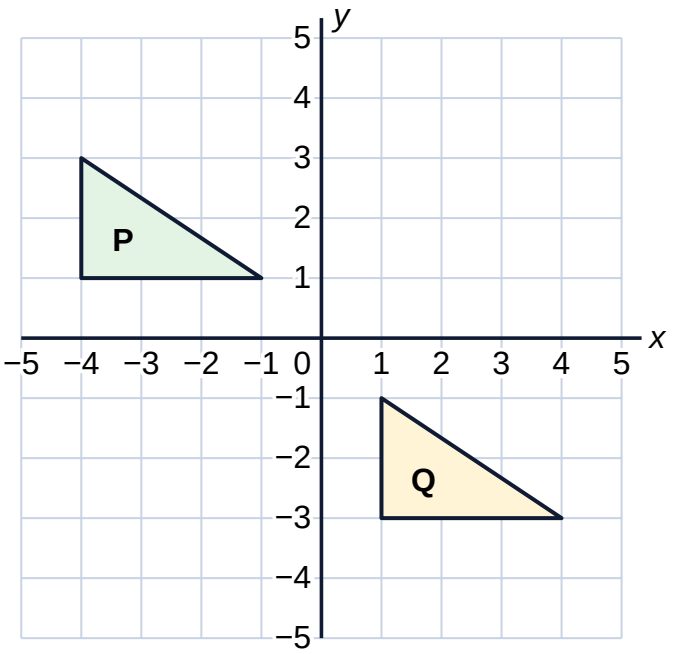
Answer: 1 mark: angle drawn between 123° and 127° AND named obtuse. Do not accept a reflex angle of 235° or the type missing.
Notes / working: Draw a straight line, put the centre of the protractor on one end, read **125°** from the scale that starts at 0° on your line, mark it, and join. 125° is between 90° and 180°, so it is an **obtuse** angle. Check: it should look a little more than a right angle plus a third of a right angle.

20 A map of the fun fair is drawn on a coordinate grid. **[1]**

(a) The ice-cream van is at (−2, 3.5). It moves 4 squares to the right and 2 squares down. Write its new coordinates.

Answer: (2, 1.5) or (2,1.5) or 2, 1.5 or (2, 1 1/2) or (2, 3/2)
Notes / working: Right 4: x = −2 + 4 = 2. Down 2: y = 3.5 − 2 = 1.5. New position **(2, 1.5)**.

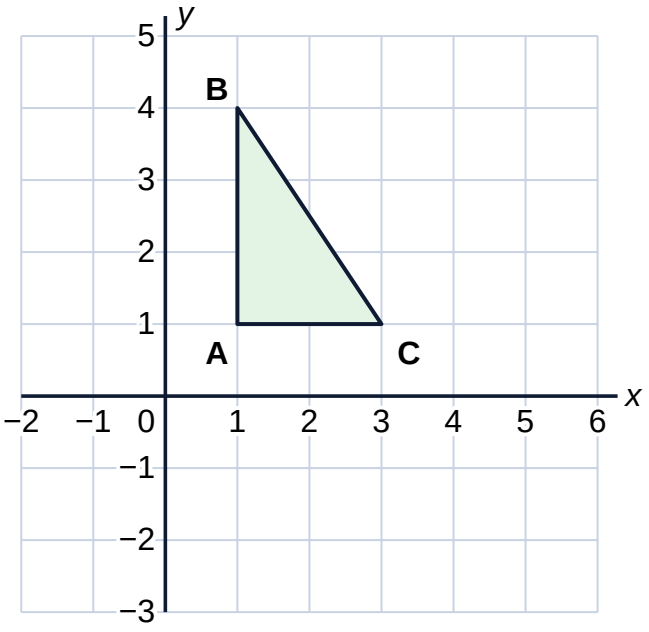
[1]



(b) Shape P is a ticket booth. It is moved to position Q.
Describe the translation from P to Q.

Answer: 1 mark: 5 (squares) right AND 4 (squares) down (accept "5 across to the right and 4 down"). Do not accept 4 right 5 down or a description of Q to P.
Notes / working: Pick one corner and follow it: the right-angle corner of P is at (−4, 1); in Q it is at (1, −3). x changes from −4 to 1 (**5 right**); y changes from 1 to −3 (**4 down**). Every corner moves the same way: translation **5 right, 4 down**.

21 The diagram shows triangle ABC on a coordinate grid. **[1]**



The triangle is rotated 90° clockwise about the vertex A.
Write the coordinates of the new position of vertex B.

Answer: (4, 1) or (4,1) or 4, 1
Notes / working: A(1, 1) stays still. B is 3 squares **straight up** from A. A quarter turn clockwise moves "up" to "right", so the new B is 3 squares **right** of A: **(4, 1)**. (C(3, 1) is 2 right of A and moves to 2 down: (1, −1).) Tip: use tracing paper and turn it about A.

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22 On a farm, h is the number of hens and d is the number of ducks. **[1]**
h − d = 12

Write two different pairs of values that h and d could have.

Answer: 1 mark: two different correct pairs of whole numbers, e.g. h = 20, d = 8 and h = 15, d = 3 (h must be 12 more than d). Do not accept negative numbers or a pair where d is more than h.
Notes / working: h − d = 12 means there are 12 more hens than ducks. h and d can **vary**: for example **h = 20, d = 8** (20 − 8 = 12) or **h = 30, d = 18** (30 − 18 = 12). Any pair where h is 12 more than d works.

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23 The rule for a sequence is: **[1]**
term = position × position + 5

Find the 12th term of the sequence.

Answer: 149
Notes / working: The terms are square numbers plus 5: 6, 9, 14, 21, … For position 12: 12 × 12 + 5 = 144 + 5 = **149**.

24 Four friends go to the bookshop sale. Together they buy 3 books at $12.80 each and 2 maps at $7.40 each. They share the total cost equally. **[2]**

How much does each friend pay? Show your working.

Answer: 13.3 $
Notes / working: Books: 3 × 12.80 = 38.40
Maps: 2 × 7.40 = 14.80
Total: 38.40 + 14.80 = 53.20
Each friend: 53.20 ÷ 4 = **$13.30**
Mark scheme: M1 for 53.2(0) seen (or 38.4(0) and 14.8(0) both seen); A1 for 13.3(0).

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25 A farm sells eggs in boxes of 12. One week the hens lay 950 eggs. **[1]**

How many boxes can be filled completely?

Answer: 79 boxes
Notes / working: 950 ÷ 12 = 79.16… So **79 full boxes** (79 × 12 = 948, with 2 eggs left over).
Common mistake: rounding up to 80 — the 80th box would not be full.